

## Can Chance Explain the Mean Difference?

AP Statistics · Unit 4 · Topic 4.2 · B: 2027-01-26 · E: 2027-01-27

### [STAR] TODAY'S PROBLEM

Suppose 50 student volunteers test a planning prompt for a logic task. They are randomly assigned to two groups of 25. Mean times are 14.2 minutes with the prompt and 17.8 with standard directions: prompt minus standard is  $-3.6$ . Under no effect, a simulation shuffles the same times into new groups. Only 18 of 2,000 shuffled differences are  $-3.6$  or lower.

**Nia:** “This is evidence that the prompt lowered mean time for these volunteers, but we could still be wrong.”

**Owen:** “The 0.009 is the probability of no effect, so there is a 99.1% chance the prompt works for every district student.”

#### Experiment results

| Group    | n  | Mean (min) |
|----------|----|------------|
| Prompt   | 25 | 14.2       |
| Standard | 25 | 17.8       |

#### FIRST TAKE — before we discuss (5 min)

Whose claim sounds more believable, Nia's or Owen's? Give one reason from the study.

#### [BOOK] ASK WHAT CHANCE WOULD DO UNDER NO EFFECT (keep on the page)

Under no treatment effect, the expected difference in means is 0. A rerandomization simulation keeps the observed responses and group sizes, randomly reassigns the responses, and repeats. The fraction at least as extreme in the claim's direction estimates how often chance alone would give such a result. It does not give the probability that no effect is true. A small fraction is evidence against chance alone, but errors remain possible. Random assignment supports causation for the study subjects; random sampling supports population claims.

Work (a)–(c) from the rules box:

**DOK 1** (a) What difference in group means would we expect if the prompt had no effect?

**DOK 2** (b) Compute  $18/2,000$ . In one sentence, explain what this proportion estimates if the prompt has no effect.

**DOK 3** (c) In 3–4 sentences, decide whose claim is better supported. Use the simulation result, explain why an error is still possible, and say why the volunteers do not represent every district student.

**Frame:** The simulation supports \_\_\_\_\_ because \_\_\_\_\_. We could still be wrong because \_\_\_\_\_.

**Frame:** The study supports a claim about \_\_\_\_\_; a district-wide claim needs \_\_\_\_\_.

**EXIT — turn this in**

One thing the rules box changed about my first take: \_\_\_\_\_